

❖ HANDS-ON 2

❖ Data Binding, Lifecycle Hooks & Component Communication

Task 1: All Four Binding Types

Goal: Practise interpolation, property binding, event binding, and two-way binding in one component.

Steps:

```
// Property binding [property] sends data from the component to the DOM.
// Two-way binding [(ngModel)] keeps the component and the DOM synchronized in both directions.
import { Component } from '@angular/core';
import { FormsModule } from '@angular/forms';

@Component({
  selector: 'app-home',
  imports: [FormsModule],
  templateUrl: './home.html',
  styleUrls: ['./home.css'],
})
export class Home {

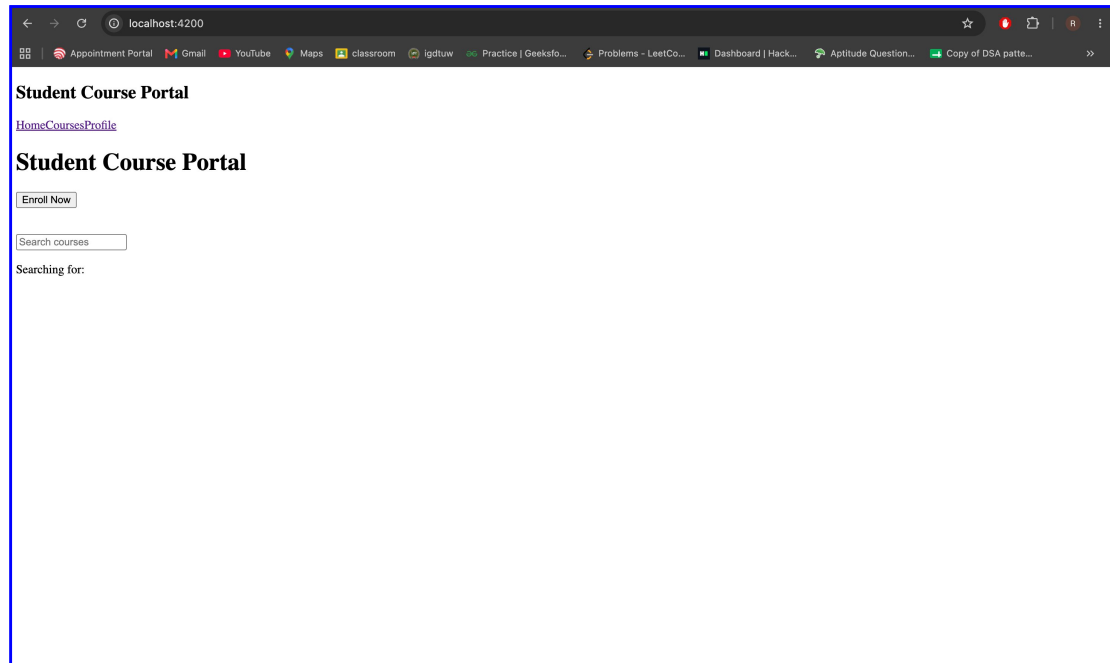
  portalName = 'Student Course Portal';

  isPortalActive = true;

  message = '';
  searchTerm = '';

  onEnrollClick() {
    this.message = 'Enrollment opened!';
  }
}
```

Task 1 result:



Task 2: Lifecycle Hooks

Goal: Implement the three most important lifecycle hooks and observe when they fire.

Task 2 result:

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Java Programming

Code: CS101

Credits: 4

Angular

Code: CS201

Credits: 3

Spring Boot

Code: CS301

Credits: 4

Angular is running in development mode.

Course input changed

Previous: undefined

Current: {id: 1, name: 'Java Programming', code: 'CS101', credits: 4}

Course input changed

Previous: undefined

Current: {id: 2, name: 'Angular', code: 'CS201', credits: 3}

Course input changed

Previous: undefined

Current: {id: 3, name: 'Spring Boot', code: 'CS301', credits: 4}

Task 3: @Input and @Output — Parent-Child Communication

Goal: Pass data down with @Input and emit events up with @Output.

Task 3 result:

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Java Programming

ID: 1

Code: CS101

Credits: 4

Enroll

Angular

ID: 2

Code: CS201

Credits: 3

Enroll

Spring Boot

ID: 3

Code: CS301

Credits: 4

Enroll

Database Systems

ID: 4

Code: CS401

Credits: 3

Enroll

Operating Systems

ID: 5

Code: CS501

Credits: 4

Enroll

Selected course ID: 5